

Lum Ko Sand

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EDUCATION

Singapore Institute of Technology – DigiPen Institute of Technology Singapore

Bachelor of Science with Honours in Computer Science in Interactive Media and Game Development
Aug 2023 – Apr 2027

Nanyang Polytechnic

Diploma in Game Development & Technology
Apr 2018 – Mar 2021

Institute of Technical Education

Higher Nitec in Games Programming & Development
Apr 2016 – Mar 2018

Nitec in Social Media & Web Development
Jan 2014 – Dec 2015

WORK EXPERIENCE

Dnasok Games – Game Developer (Self-Employed)

Oct 2021 – Feb 2026 | Singapore

- Designed, developed, tested, and deployed Unity 3D applications using C#, following a full software development lifecycle.
- Built modular systems and reusable components, improving maintainability and performance across multiple releases.
- Managed version control, builds, and releases independently, demonstrating ownership and accountability.

VizioFly (Virtual Reality Singapore) – Software Engineer Intern

Jun 2020 – Aug 2020 | Singapore

- Developed immersive AR/VR and simulation-based applications using Unity and C#, working in an agile, project-based environment.
- Collaborated with cross-functional teams to deliver client-facing solutions on schedule.
- Designed responsive user interfaces with cross-platform compatibility considerations.

Viswire – Software Engineer Intern

Oct 2017 – Dec 2017 | Singapore

- Contributed to real-time simulation systems used for professional training and research.
 - Worked with engineers and researchers to implement features and resolve technical issues.
 - Produced technical documentation to support system usability and maintenance.
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RELEVANT PROJECTS

Ukemochi (Custom C++ Engine) – Student Project | Technical Lead

- Led technical development of a custom C++ engine, designing core systems such as physics, collision detection, and UI frameworks.
- Applied object-oriented design and modular architecture principles to build scalable systems.
- Collaborated with an 8-member team, supporting code integration and technical decision-making.

Tiny Arcade Racers (Indie Racing Game) – Personal Project

- Delivered a complete software product from concept to deployment across PC and mobile platforms.
- Optimised performance and resource usage for different runtime environments.
- Implemented user-focused features based on testing and feedback.

NATIONAL SERVICE

Singapore Police Force – Protective Security Command Trooper

May 2021 – May 2023 | Singapore

- Conducted patrols and safeguarded key installations at security-sensitive sites.
- Secured Critical Infrastructure and supported major national events.
- Awarded Best in Operational Fitness and developed leadership, teamwork, and communication skills in high-pressure environments.

AWARDS AND CO-CURRICULAR ACTIVITIES

- Distinction in Business for Digital Media Industry and in Principles of Game Design.
- Director's List recipient at Institute of Technical Education.
- Active participation in student game clubs and competitions, providing guidance to members.

SKILLS

Programming & Development

- C++, C#, scripting
- Object-Oriented Programming (OOP)
- Gameplay & frontend feature development
- Debugging, testing & performance optimisation

Tools, Systems & Platforms

- Application architecture & modular design
- Cross-platform development (Windows, Mobile)
- Version control (Git), Visual Studio
- Unity 3D (Gameplay, Physics, UI, Animation, Asset Pipeline, VR/AR)

Soft Skills: Structured problem-solving, technical communication & documentation, team collaboration (designers, artists, QA)